Philosophy of Science
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Lecture 4

Explanations and Causes

Motivation of this lecture

- In International Studies, you will be challenged to…
  - explain outcomes and observations
  - pinpoint the causes of events
- This lecture familiarises you with…
  - some classic accounts of explanation
  - concept of causal explanation
- Helping you to become effective and reflective scientists

Plan of this lecture

- What is explanation?
  - Explanation, knowledge, and truth
- Explanation as deduction
  - Two classic models of explanation
  - Proposed by Carl G. Hempel
- Objections to these
  - … and a move to new causal approaches to explanation

Reading for this lecture

- Michael Strevens, “Scientific Explanation”
  - In *Encyclopedia of Philosophy*, edited by Donald M. Borchert, second edition
  - Macmillan Reference, 2006
- Available via Brightspace

Nomothetic vs. idiographic again

- Nomothetic approach
  - Seeks causes and formulates explanations
  - "Explanation" is nomothetic terminology
- Idiographic approach
  - Mistrusts concepts of "explanation" and "cause" as mechanistic, reductionist
  - Prefers "interpretation" and "understanding"

Knowledge and explanation

- Suppose we had a record of all events
  - Would science then be complete?
- No. We would still lack:
  - Awareness of connections between events
  - Knowledge of which events were determined and which accidental
  - Ability to intervene effectively in the world
- Explanations deliver these further attainments
  - A good explanation can be revelatory

Explanation and truth

- For an account to have the status of explanation, it must be true
  - Most philosophers agree
  - Otherwise, we have a pseudo-explanation
- Example: mythological accounts
  - "Lightning occurs because Zeus is angry"
  - This account fails to explain lightning
  - Reason: it is not true

What is an explanation?

- An answer to a "why" question
  - Account that allows us to grasp why things are as they are
- Connection between…
  - *Explanandum*: what is to be explained
  - *Explanans*: what does the explaining
- For further detail…
  - We need to construct philosophical models of explanation

Carl G. Hempel, 1905–1997

- Born in Germany
- PhD in philosophy, Berlin, 1934
- Worked at Princeton and Pittsburgh universities
- Representative of logical empiricism

Hempel's models of explanation

- Deductive-nomological (DN) model
  - Famous and influential account
  - Proposed by Hempel in 1965
  - Much criticised but still a basis of discussion
- Inductive-statistical (IS) model
  - Extension of DN model to probabilistic phenomena
- These models…
  - offer a view of which formal requirements an explanation must satisfy

Deductive-nomological model

- On the DN model, an explanation is:
  - a valid deductive argument (D)
  - from true premises
  - that include at least one scientific law (N) and descriptions of some particular facts
  - to a conclusion that states the *explanandum*
- Influential approach
  - Explanation as a form of deduction

DN model: logical structure

- A DN explanation is a valid deductive argument of the following form:

$L_1, \ldots L_n$	Laws or true generalisations
$C_1, \ldots C_n$	Descriptions of particular facts
————	
E	Statement of the *explanandum*

DN model: example

- Why did trade volume fall in case *C*?

  When trade barriers rise sharply, trade volume typically falls

  In case *C*, trade barriers rose sharply

  ————————————————————

  So, in case *C*, trade volume fell

- Gives us a degree of understanding of the *explanandum*

DN model and prediction

- The deductive argument…
  - that constitutes an explanation according to the DN model…
  - is also what we use to make a prediction
  - Example: we can use it to predict that trade volume will fall at a future time
- Parallel of explanation and prediction
  - One difference: the premises of an explanation must be true (recall sheet 7)
  - A successful prediction, by contrast, may follow from false premises

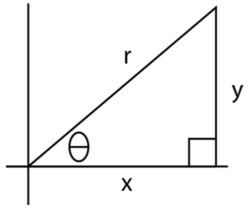
DN model: critical scrutiny

- We have pre-existing intuitions about good and poor explanations
  - Does the DN model satisfy them?
- For instance, we want a model of explanation to rule out explaining…
  - a cause on the basis of its effect
  - an event on the basis of irrelevant information
- The DN model allows both these things
  - It is thus too permissive

Laws, causes, and effects

- Typical laws do not distinguish causes and effects
- In science: time-symmetric equations
- Example: $F = Gm_1m_2/r^2$
  - Enables us to deduce the state of the solar system at time t_2 from its state at t_1—going forwards in time
  - But also to deduce its state at t_1 from its state at t_2—going backwards in time
- Everyday example: flagpole / shadow

Flagpole / shadow



θ: angle of sun
y: height of flagpole
x: length of shadow

- Trigonometric relation $\tan \theta = y / x$
- From θ and y, we can deduce x
- Equally, from θ and x, we can deduce y
- The DN model allows both as explanations

DN model: objection 1

- Plugging a law into the DN model will thus allow us to "explain"…
  - an effect by reference to its cause
  - but also a cause by reference to its effect
- However, we do not think that an effect explains its cause
  - Shadow cannot explain height of flagpole
- Flaw of the DN model: too permissive
  - By relying on laws, it lacks the resources to distinguish causes and effects

DN model: objection 2

- We think that an outcome is explained by relevant information
  - Irrelevant information provides no understanding
- Yet the DN model accepts as a valid explanation some accounts that cite irrelevant factors
  - Further flaw of the DN model
  - Classic example: hexed salt

Example: hexed salt

- Why did this salt dissolve in water? Apply the DN model:

  All hexed salt dissolves in water
  This salt was hexed
  ———————————————————
  Hence, this salt dissolved in water

- But all salt dissolves in water
  - This deductive argument gives us no understanding of the *explanandum*

Inductive-statistical model

- Extension of DN model to probabilistic situations
  - Cases in which $0 < p < 1$, where p is the probability of an outcome
  - Proposed by Hempel, 1965
- On the IS model, an explanation is…
  - an argument that establishes that the *explanandum* had high probability
  - We call this model "inductive" because it uses probable, not certain, reasoning

IS model: example

- *Explanandum*: death of plant *B*
  - Why did *B* die?
- Explanation of IS form:

  Any plant sprayed with defoliant *D* has probability 0.8 of dying
  Plant *B* was sprayed with *D*
  ——————————————————————————
  Therefore, *B* died (probability 0.8)

- As before, the explanation yields some understanding of the *explanandum*

IS model: counterexample

- Some good scientific explanations…
  - do not make the *explanandum* highly likely
  - showing that the IS model is too restrictive
- Example: why did patient *S* develop paresis?
  - Because patient *S* had untreated syphilis
  - This is a good explanation: only patients with untreated syphilis develop paresis
  - But such patients have only probability 0.3 of developing paresis—not high

Hempel's approach: critique

- Hempel thought that…
  - laws were enough to capture causality
  - That is why he placed laws at the heart of the DN model
- But counterexamples to the DN model reveal that a factor can be…
  - part of the sufficient condition for the occurrence of an outcome
  - without being a cause of that outcome

From DN to causal models

- Conclusion: explanations should…
  - track causes
  - not merely state sufficient conditions for occurrence
- DN model fails this requirement
  - We need more causal models of explanation
- Example: causal-mechanical model
  - Proposed by Wesley C. Salmon, US philosopher of science, 1984

Causal-mechanical model 1

- An explanation of event *E* is…
  - a description of (part of) the causal interactions and causal processes that led to *E*
- Two key concepts of this account:
  - Causal interaction and causal process
- Illustration: game of cricket

Ellyse Perry, Australia all-rounder

Causal-mechanical model 2

- Causal interaction:
  - Spatio-temporal intersection of two causal processes that modifies both
  - Example: stroke of a cricket bat that leaves a scuff on the ball
- Causal process:
  - Physical process able to transmit a mark in a continuous way
  - Example: movement through space of a cricket ball bearing a scuff

Further models of explanation

- There are other accounts of explanation
  - Beyond the scope of this lecture
- Examples:
  - Unification model: to explain *E* is to reduce *E* and similar outcomes to a general class or pattern—extension of Hempel’s DN model
  - Functional explanation: to explain *E* is to show that *E* contributes to a goal *G*
  - Narrative explanation: to explain *E* is to tell a story that leads to *E*

Recapitulation

- Carl G. Hempel’s classic models of explanation
  - Deductive-nomological model
  - Inductive-statistical model
- Criticism of these models
  - Failure to track causal relations
- New causal approaches to explanation